SIH25077

Unauthorized Electric Fence Detection System

Full Technical & Functional Documentation

Document Scope

This document consolidates the complete hardware architecture, software architecture, end-to-end data flow, and dashboard specification (fleet-level and per-meter) for the Unauthorized Electric Fence Detection System — a low-cost IoT + ML solution to detect and alert authorities to illegal direct-mains fence connections before they cause loss of human or animal life.

Version 1.0 | August 2026

1. Executive Summary

Villages bordering forest land often connect farm-boundary fences directly to 230V grid mains, instead of using a legal low-voltage energizer, to deter wildlife from crops. A legal energizer delivers a brief, non-lethal pulse; a direct mains tap is continuously live and lethal — killing elephants, cattle, and people every year across states such as Kerala and Assam. Today there is no automated way to detect an illegal connection before it kills.

This project places a low-cost current-sensing device on the electricity supply line feeding a fence (at the drop-line or transformer, not the private fence itself), continuously analyzes the electrical waveform, classifies it as a legal energizer, an illegal direct tap, or an ordinary household appliance, and alerts forest/electricity officials with GPS location in real time — before an animal or person is harmed.

The system has three layers: a sensing/hardware layer, a signal-processing and ML software pipeline, and a two-tier web dashboard (fleet overview + per-meter detail) used by officials to monitor, verify, and act on alerts.

2. Problem Statement

- Illegal direct-mains fencing is a continuously live, lethal hazard — unlike legal energizers which are designed to be safe (IS/IEC 60335-2-76).
- There is no existing detection mechanism; violations are discovered only after an animal or human death, or through manual patrols and public tip-offs.
- Forest and electricity departments lack real-time, location-tagged visibility into where illegal taps exist across a large, remote geographic area.
- Remote forest-boundary regions frequently lack reliable WiFi/cellular coverage, complicating any always-online IoT solution.

3. Core Detection Insight

Legal energizers and illegal grid taps are electrically distinguishable at the supply meter. This measurable difference is the foundation of the entire system:

Parameter	Legal Energizer	Illegal Direct Grid Tap
Waveform	Short-duration pulses	Continuous 50Hz sine wave
Repetition	~1 pulse per second (0.8-1.2s)	50Hz continuously
Duty Cycle	Very low (<5%, typically <1%)	Very high (~100%)
Pulse Width	< 0.1s (very short)	Continuous
RMS Current	Low (only during charging)	Higher (continuous load/leakage)
Harmonic Content	Low-frequency pulses, low 50Hz component	Strong 50Hz component with harmonics (100Hz, 150Hz...)

Parameter	Legal Energizer	Illegal Direct Grid Tap
THD (Total Harmonic Distortion)	< 5%	> 25% (with appliance load-mixing on same line)
Danger Level	Designed safe, startling but non-lethal	Very dangerous / illegal / potentially lethal

Why this matters:
Because the difference shows up clearly in both the time domain (waveform shape) and frequency domain (FFT spectrum), the detection logic can be validated with simple threshold rules first, then hardened with a trained ML classifier — giving the team both a working demo and a defensible technical differentiator.

4. System Architecture — Three Layers

The system is organized into three layers that map directly to the sections that follow:

1. Hardware Layer — sensing, signal conditioning, processing, communication, and power, physically installed at the fence supply point.
2. Software Layer — the 9-stage pipeline that ingests raw telemetry, extracts features, classifies events, stores data, and issues alerts.
3. Dashboard Layer — two tiers of web interface used by forest/electricity officials: a fleet-wide overview and a per-meter detail view, both backed by a Device Registry & Access Control layer.

5. Hardware Architecture

The device is installed near the energy meter on the drop-line feeding the fence, so it observes the exact current the fence line draws without needing to touch the live fence wire itself.

5.1 Sensing & Input

Component	Function
CT (Current Transformer) Clamp Sensor	Non-intrusive current sensing — clamps around the live wire, reads current flow without direct electrical contact.
Voltage Divider	Steps down mains voltage sample for reference/synchronization with the current waveform.
Aux Sensors (Temp / Humidity / Gas)	Optional environmental monitoring for enclosure health and outdoor deployment conditions.
Tamper Switch (Reed / Micro-switch)	Detects physical removal or opening of the enclosure; triggers a separate tamper alert.

5.2 Signal Conditioning

Raw CT output is small and noisy; it must be conditioned before the microcontroller's ADC can sample it accurately.

- Anti-Aliasing Filter — removes high-frequency noise above the Nyquist limit before sampling.
- Amplifier — boosts the millivolt-level CT signal to a usable ADC input range.
- Isolation — galvanically isolates the low-voltage electronics from the sensed mains-adjacent line for user and board safety.
- ADC (Analog-to-Digital Converter) — digitizes the conditioned analog waveform at a sampling rate sufficient to resolve harmonics up to at least the 5th order (250Hz) of the 50Hz fundamental.

5.3 Processing & Control

- ESP32 Microcontroller — dual-core Wi-Fi/BLE SoC; runs sampling, feature extraction, local classification logic, and manages all peripherals.
- SD Card Module — local buffering of raw/processed telemetry when connectivity is unavailable, preventing data loss in low-coverage zones.
- RTC (Real-Time Clock) — maintains accurate timestamps for events even through power loss, critical for violation-history logging and legal evidentiary use.

5.4 Communication

Link	When Used
Wi-Fi (Internet)	Primary link where local internet infrastructure exists (near towns/substations).
LoRa Module	Long-range, low-power backup for remote forest zones with no cellular/Wi-Fi coverage; also used for mesh relay between nearby units.
GSM Module	Cellular fallback for SMS-based alerting where mobile network exists but Wi-Fi/LoRa gateway does not.

Only one link is required per deployment; the choice depends on the coverage available at that specific installation site (see Section 8 — Data Flow, Step 4).

5.5 Power Supply

- Solar Panel — primary power source for remote, off-grid installation points.
- Battery (7.4V / 12V) — stores solar energy and powers the unit through night/low-sun periods.
- Power Management (DC-DC, LDO) — regulates battery voltage to stable rails for the ESP32 and sensors.
- Protection (Fuse, TVS diodes) — guards against surges, reverse polarity, and short-circuit damage.

5.6 Output / Actuation & Indication

- Relay / Contactor — optional remote power cut-off capability at the supply point (used only where policy/legal authority permits automatic disconnection).
- Buzzer — local audible alert for field technicians during installation/testing.
- Status LED — quick visual indication of device health (powered / connected / alert state).

5.7 Bill of Materials & Cost

Component	Approx. Cost (₹)
Clamp-on current sensor (CT)	200 - 400
ESP32 Dev Board	300 - 400
GPS module	300 - 500
GSM / LoRa module	400 - 600
Enclosure, wiring, battery	300 - 400
Total per unit	~1,500 - 2,300

6. Software Architecture

Telemetry from the hardware layer flows through a 9-stage software pipeline before reaching the dashboard. Each stage is a discrete, independently upgradeable module.

#	Stage	Tech / Method	What it does
1	Data Ingestion	pyserial, MQTT, HTTP REST	Reads high-frequency raw telemetry via USB/UART locally, and receives wireless sensor data (Wi-Fi/LoRa/GSM) from field units.
2	Signal Processing	scipy.signal, numpy.fft	Filtering, windowing, peak/envelope detection, and FFT to extract clean spectral information from the raw waveform.
3	Feature Extraction	NumPy, Pandas	Computes RMS, Peak, Crest Factor, Pulse Width/Rate, Duty Cycle, Power Factor, THD, and other discriminative features.
4	Machine Learning & AI	scikit-learn, Joblib / ONNX	Classifies the electrical event (legal energizer / illegal tap / appliance) using RF, SVM, DT, or kNN models tuned for high recall, low latency.
5	Backend & APIs	FastAPI, WebSockets	Asynchronous REST API and WebSocket server that serves real-time predictions and telemetry to the dashboard.
6	Data Storage & Logging	SQLite / Time-Series Store, Pandas	Stores time-series telemetry, anomaly logs, ML predictions, and all system events with timestamps.
7	Visualization & Analytics	Matplotlib / Interactive Frontend	Real-time waveform and FFT plots, harmonic analysis, and live alert dashboards (see Section 9).
8	Domain Framework	NILM Principles	Non-Intrusive Load Monitoring: load disaggregation, transient signature recognition,

#	Stage	Tech / Method	What it does
			power-quality analysis to separate appliances from fence taps.
9	Alerts & Notifications	Dashboard / SMS / Email / Push	Sends real-time alerts to officials via dashboard, SMS, email, or push notification the moment a violation is confirmed.

A Model Feedback Loop connects Stage 9 back to Stage 4: officials confirming or dismissing an alert on the dashboard feeds labelled examples back into the ML pipeline, continuously improving classification accuracy — turning every real deployment into additional training data (see Section 12).

7. Device Registry & Access Control

This is the connective layer between the hardware/software pipeline and the dashboard — it did not exist as an explicit block in the original architecture diagram and is added here as it is required for any multi-device, multi-official deployment.

7.1 Device Metadata (per unit)

Field	Description
Device ID	Unique hardware identifier (burned-in serial / MAC-derived ID).
GPS Coordinates	Fixed install-point latitude/longitude, captured at commissioning.
Zone / Transformer ID	Which transformer, drop-line, or forest-risk zone this unit covers.
Gateway Mapping	Which LoRa gateway / cell tower / Wi-Fi AP this device reports through — needed because one gateway can relay many field units.
Install Date & Technician	Commissioning record for maintenance and audit trail.
Firmware Version	For staged OTA rollout and vulnerability tracking.
Owning Authority	Forest Department / State Electricity Board / Panchayat — determines which officials see this device by default.

7.2 User Roles & Access Control

Role	Access
Super Admin	Full access — device provisioning, user management, system configuration, all zones.
Forest Department Official	View/act on alerts within their assigned zone(s); mark verified/resolved; cannot edit device configuration.

Role	Access
Electricity Board Official	View/act on device-health and tamper alerts within their jurisdiction; manage physical disconnection workflow.
Field Technician	Device commissioning, firmware updates, physical maintenance logging; limited dashboard read access.
Viewer / Auditor	Read-only access to historical data and reports, e.g. for policy or funding review.

- Authentication: device-to-backend calls are authenticated (per-device API key / signed payload) so a spoofed device cannot inject fake alerts.
- Transmission is encrypted in transit (TLS for Wi-Fi/GSM links; LoRaWAN payload encryption for LoRa links).

8. End-to-End Data Flow

This section explains, step by step, exactly how a reading travels from the physical fence wire to an official's screen and back to a resolved action.

1. Sense: The CT clamp sensor continuously measures current flowing through the fence supply line, without breaking or touching the live wire.
2. Condition & Digitize: The analog signal is filtered, amplified, isolated, and converted to digital samples by the ADC at a rate fast enough to resolve harmonics.
3. Local Compute: The ESP32 runs signal processing (FFT) and feature extraction (duty cycle, RMS, THD, crest factor) on-device in real time.
4. Local Classification: A lightweight on-device model (or threshold rule as a fallback) produces a first-pass classification: SAFE (legal energizer or normal appliance) or ALERT (illegal tap signature).
5. Transmit: The device sends a compact telemetry packet — device ID, timestamp (from RTC), classification, key features, GPS — over whichever link is available at that install site: Wi-Fi (if internet present) → else LoRa (to a nearby gateway) → else GSM (SMS/data fallback). If no link is available, data buffers locally to the SD card and syncs when connectivity returns.
6. Ingest: The backend's Data Ingestion stage (MQTT/HTTP REST) receives the packet and hands it to the processing pipeline.
7. Server-Side Verification: The full ML pipeline (Stages 2-4) re-validates the classification server-side using the complete feature set and the NILM appliance-vs-fence model, since the server can run a heavier model than the ESP32 can.
8. Persist: The confirmed event, along with raw/processed waveform data, is written to the time-series data store (Stage 6), tagged with the Device ID and Zone from the Device Registry.
9. Broadcast: The Backend & APIs stage pushes the update over WebSocket to any dashboard currently open, and evaluates alert rules.
10. Alert: If the event is classified ALERT (illegal tap) or TAMPER (enclosure opened), Stage 9 immediately fires notifications — dashboard banner, SMS, and/or email/push — to every official whose role and zone match that device (per Section 7.2).
11. Act: The receiving official opens the Per-Meter Dashboard (Section 9.2) for that device, reviews the live waveform/FFT and history, dispatches a team to physically verify and disconnect the illegal tap, and marks the alert Verified / Resolved / False Positive on the dashboard.
12. Feedback Loop: The official's resolution label is stored alongside the original waveform data as a new labelled training example, feeding back into Stage 4 to improve future classification accuracy (Section 12).

9. Dashboard Specification (Two-Tier)

Officials interact with the system through two linked views: a fleet-wide overview for situational awareness across all deployed units, and a per-meter detail view for investigating and acting on a specific device.

9.1 Tier 1 — Fleet / Overview Dashboard

Purpose: give an official a single glance at the health and alert status of every device in their jurisdiction, and let them drill down into any one of them.

Map View

- Interactive map (e.g. Leaflet/Mapbox) plotting every device at its registered GPS coordinate.
- Marker color-coded by current status: Green = Safe, Red = Active Alert (illegal tap), Orange = Tamper Detected, Grey = Offline.
- Clicking a marker opens a summary popup with a link into that device's Tier 2 detail page.

Fleet KPI Cards

KPI	What it shows
Total Devices Deployed	Count of all registered units in the selected zone/filter.
Active Alerts	Count of devices currently in ALERT (illegal-tap) status.
Devices Offline	Count of units that have missed their expected check-in interval.
Tamper Events (24h)	Count of enclosure-tamper triggers in the last 24 hours.
Average Battery / Solar Health	Fleet-wide power health indicator.
Verified Violations (This Month)	Running count of confirmed, officially-actioned illegal taps.

Live Alert Feed / Table

A sortable, filterable table of recent events, each row showing:

- Timestamp
- Device ID
- Zone / Transformer
- Event Type (Illegal Tap / Tamper / Device Offline)
- Status (New / Acknowledged / Team Dispatched / Resolved / False Positive)
- Assigned Official

Filters & Controls

- Filter by zone, owning authority, device status, date range, and event type.
- Export filtered view as CSV/PDF for reporting to forest/electricity department leadership.

9.2 Tier 2 — Per-Meter Detail Dashboard

Purpose: everything an official needs to investigate one specific device's current state and history in depth, and take action on it.

Header / Identity Panel

- Device ID, GPS coordinates (with mini-map), Zone/Transformer ID, Owning Authority, Install Date.
- Large status badge: SAFE / ALERT / TAMPER / OFFLINE.

Live Signal Panels

- Real-time Time-Domain Waveform — current (A) vs time (ms), refreshed live via WebSocket.
- Real-time FFT Spectrum — magnitude vs frequency (Hz), highlighting the 50Hz fundamental and harmonics (100Hz, 150Hz...).

Key Electrical Metrics (live values)

Metric	Typical Safe Range	Typical Alert Range
RMS Current	~2.5 A (household mix)	~7.5 A (continuous fence load)
Peak Current	~6 A	~18 A
Crest Factor (Peak/RMS)	~2.2	> 3.0
THD (Total Harmonic Distortion)	< 5%	> 25%
Duty Cycle	< 5% (energizer) or matches known appliance profile	~100% continuous
Dominant Frequency	50 Hz, clean	50 Hz, distorted with strong harmonics

Historical Trend Charts

- Time-series graphs of RMS current, THD, and duty cycle over selectable windows (24h / 7d / 30d).
- Overlay markers for past alert and tamper events on the timeline.

Device Health Panel

Field	Description
Battery %	Current charge level of the on-site battery.
Signal Strength	Wi-Fi/LoRa/GSM link quality (RSSI).
Last Seen	Timestamp of most recent successful check-in.
Firmware Version	Currently running firmware build.
SD Buffer Status	Whether locally buffered data is pending upload.

Event / Violation History Log

- Chronological log of every ALERT and TAMPER event for this device, each with timestamp, duration, resolving official, and outcome (Verified / False Positive / Resolved).
- Downloadable as a report — usable as a field-verification and enforcement record.

Action Panel

- Acknowledge Alert — stops repeat notifications while investigation is underway.
- Dispatch Team — logs which field team was sent and when.
- Mark Verified / Resolved / False Positive — closes the event and feeds the Model Feedback Loop (Section 6, Section 12).
- Add Note — free-text field for on-site findings.

10. Alert Classification & Notification Logic

Decision flow used by both the on-device fallback logic and the server-side model:

1. Acquire current signal from the CT sensor at the meter.
2. Reconstruct the waveform from raw ADC samples.
3. Extract features: duty cycle, pulse width, frequency, RMS, THD, crest factor.
4. Compare against thresholds and/or the trained ML model.
5. If the signature matches the legal-energizer profile → status SAFE / LEGAL, no alert.
6. If the signature matches the direct-power/illegal-tap profile → status ALERT / ILLEGAL, notification sent to Mobile App, Authority/Control Room, and Cloud Dashboard.
7. If the signature is ambiguous (could be an appliance) → routed through the NILM appliance-classifier (Stage 8) before a final decision is made, to minimize false positives.

11. Handling the Appliance vs. Fence Ambiguity

At a shared supply point, continuous current could come from a fence tap or from a normal appliance (pump, fridge, heater). This is addressed with an AI/ML-based load-signature classification approach (Non-Intrusive Load Monitoring):

- A lightweight model (Random Forest / small 1D-CNN) is trained on waveform features — duty cycle, harmonic distortion, signal steadiness, startup transients — to distinguish appliances from a flat, featureless direct-tap signal.
- Initial training uses public appliance datasets (REDD, PLAID, WHITED) combined with a self-generated fence-signal dataset from controlled low-voltage mock circuits.
- Placement strategy also reduces ambiguity structurally: installing directly on the fence supply drop-line (rather than a shared household line) removes most of the ambiguity by design.
- Every official-confirmed alert becomes new labelled training data, continuously improving the classifier as the fleet scales (Model Feedback Loop, Section 6).

12. Rollout Plan

1. Phase 1: Deploy at drop-lines/transformers in documented high-risk zones, using existing forest department electrocution data (e.g. Kerala, Assam).
2. Phase 2: Expand coverage guided by real alert patterns and hotspot data generated by the system itself.
3. Future Scope: Wider transformer-level deployment using the mature ML classifier, refined with real mixed-load field data.

13. Impact

- Prevents deaths before they happen, instead of reacting after the fact.
- Removes reliance on manual patrols and public tip-offs.
- Gives officials exact, actionable GPS locations for verification and disconnection.
- Scales efficiently — one unit at a transformer/drop-line can cover dozens of farms fed by the same source.
- Builds a violation-history dataset that guides future enforcement and patrol priorities.

14. Anticipated Questions & Answers

"What if someone removes the device?"

It is installed on utility infrastructure (drop-line/transformer), not private property, and is backed by a tamper-detection switch that fires a distinct alert if the enclosure is opened or removed.

"How do you separate fence current from appliance current?"

Fence-line-specific placement removes most ambiguity by design. For wider transformer-level coverage where lines are shared, the NILM-based ML load-signature classifier (Section 11) distinguishes appliances from a flat direct-tap signature.

"Can this cover all of India instantly?"

No — a phased rollout starting with documented high-risk zones, expanding based on real alert data generated by the system itself (Section 12).

"What about network coverage in remote forests?"

The communication layer falls back automatically: Wi-Fi where available, LoRa for long-range low-power backup, GSM/SMS as a cellular fallback, and local SD buffering when no link is available at all (Section 8, Step 5).

"Is the demo safe?"

Yes — the demo uses two low-voltage (9-12V) mock circuits, one pulsing once per second (simulating a legal energizer) and one continuous (simulating an illegal tap), so the waveform physics is demonstrated without any real mains-voltage hazard on stage.

15. Glossary

Term	Meaning
CT Sensor	Current Transformer — clamp-on sensor that measures current without direct electrical contact.
THD	Total Harmonic Distortion — measure of waveform distortion relative to a pure sine wave.
Duty Cycle	Fraction of time a signal is 'on' within a repeating cycle.
NILM	Non-Intrusive Load Monitoring — technique for identifying individual loads from aggregate electrical signals.
FFT	Fast Fourier Transform — converts a time-domain signal into its frequency-domain spectrum.
RTC	Real-Time Clock — hardware clock that keeps accurate time independent of the main processor/power.
LoRa	Long Range — low-power, long-distance wireless communication protocol.
OTA	Over-The-Air — remote firmware update delivery method.